Cache Admission Study Note

1. Admission and measurement

Study ID CA-09 identifies this synthetic cache experiment.

The cache capacity is 64 MiB.

The observation window recorded 800 hits and 200 misses.

Equation 1: hit ratio = hits / (hits + misses).

The observed hit ratio is 800 / 1000 = 0.80.

Admission rejects any object larger than 8 MiB.

Eviction occurs when admitting an object would exceed capacity.

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2. Synthetic policy comparison

Table 1. Policy | hit ratio | median lookup latency

FIFO | 0.71 | 12 ms

LRU | 0.80 | 9 ms

TinyLFU | 0.84 | 11 ms

All table values are synthetic and are not production measurements.

Admitted objects have a time-to-live of 600 seconds.

This note does not measure energy consumption.